

Rajath Sathyanarayana

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY

Master of Science in Quantitative and Computational Finance

Atlanta, Georgia

Jan 2025-May 2026

- GPA: 3.75
- Key Coursework: Data Mining and Statistical Learning, Derivative Securities, Fixed Income, Systems for Computational Finance, Numerical Methods in Finance, Stochastic Processes, International Finance, Valuations and Investments, Time Series and Regression Analysis, Financial Optimization.

CARNEGIE MELLON UNIVERSITY, Language Technologies Institute

Pittsburgh, Pennsylvania

Graduate Certificate in Machine Learning and Data Science

Dec 2023-Aug 2024

- GPA: 4.0
- Key Coursework: LLMs, NLP, ML solution design, optimization and deployment, data visualization and performance analysis.

PES UNIVERSITY

Bangalore, India

Bachelor's in Computer Science and Engineering with specialization in Data Science

Aug 2017-May 2021

- GPA: 8.7/10

EXPERIENCE

FISERV

Atlanta, USA

Data Science Capstone Intern

Jan 2026-May 2026

- Partnered with Fiserv to help leverage their small business index to forecast Personal Consumption Expenditure.
- Built OLS, ARMA and heteroskedastic statistical models to predict rate of change in expenditure based on macro and index related features; obtained 75 percent plus directional accuracy.

AMAZON

Bangalore, India

Software Development Engineer - Data & Analytics

Nov 2022-Dec 2024

- Spearheaded performance reporting data pipelines and analytics dashboards for advertisers to analyze their campaign performance and granular reporting on a \$16B+ revenue-generating platform.
- Introduced programmatic API access for association of creatives and added currency and language of preference for users on the DSP portal.
- Enabled moderation teams to view top level categories of browse-nodes and ASINs which enabled a ~7x decrease in man hour spent on policy checks and implemented pre-moderation checks for restricting certain Ad sizes which need disclaimers.

COUCHBASE - NoSQL Database

Bangalore, India

Software Engineer

Jan 2021-Nov 2022

- Consulted and solved customer issues worldwide which entails understanding the nature of highly distributed, highly available and low latency database systems. Developed internal log parsing and monitoring tools to accelerate root-cause analysis for support and engineering teams.

SBX CORP

Remote

Machine Learning Research Intern

Jan 2020-July 2020

- Constructed a diagram learner which converts a biological pathway (with entities and relationships) into a machine-readable format leveraging computer vision and deep learning techniques.

GE HEALTHCARE

Bangalore, India

Machine Learning Intern

Jun 2019-Aug 2019

- Prototyped an automatic organ volume estimator from ultrasound images using segmentation, 3D visualization and volume approximation to support faster, more accurate diagnosis in primary healthcare.

RELEVANT PROJECTS

Statistical Arbitrage, Factor Modeling and Mean Variance Portfolio Optimization

Fall 2025

- Built a systematic market-neutral PCA based trading strategy. Modeled the idiosyncratic returns as a mean-reverting process.
- Estimated factor redundancies through regression and tested the abilities of Fama French factors to explain returns on test assets.
- Constructed the mean-variance efficient frontier, using historical returns and correlations.

Quantitative Research, Live Trading and Back Testing Engine

Summer 2025

- Selected equities with strong revenue and net income growth, fair valuations, and sustained positive operating cash flow; and validated outperformance using statistical tests. The hypothesis delivered 25.4% CAGR, Sharpe 1.14, over a 10-year period, outperforming broader markets.
- Created a modular live trading and back-testing engine with near real-time tick ingestion and bar aggregation through Interactive Brokers API integration which supported customizable execution frequencies across multiple asset classes.

Credit Card Fraud Detection

Spring 2025

- Developed a machine learning pipeline to detect fraudulent credit card transactions. Included data preprocessing, EDA, feature engineering, model selection (XGBoost, Logistic Regression, SVM), and optimized for a custom utility-based objective.

PUBLICATIONS

- Rajath, S., et al. "Data Mining Tool to Help the Scientific Community Develop Answers to Covid-19 Queries." 2021 Fifth International Conference On Intelligent Computing in Data Sciences (ICDS). IEEE, 2021. [Link](#)
- Rajath, S., Sumukh K. Aithal, and Natarajan Subramanyam. "Domain Shift in Capsule Networks." ICPRAM. 2021. [Link](#)
- Arun, A., Murali, V. S., Rajath, S., & Nagpal, R. "HiPAES: A Novel High Performance Parallel AES." In 2020 International Conference on Advanced Computer Science and Information Systems (ICACSIS) (pp. 263-268). IEEE. [Link](#)

SKILLS

Programming Languages: Python, Java, C++, C, SQL

Libraries: Pytorch, TensorFlow, Matplotlib, Seaborn, Pandas, NumPy, Spring

Certifications: Machine Learning from Stanford University (Coursera), Deep Learning Specialization (Coursera), Business Fundamentals Offered by Wharton (Coursera)